Assess whether a MIR ATR method can measure water content in a petroleum line, with the analyzer in another building reusing an existing silica fiber cable.

DECISION STATUS **Provisional feasibility assessment**

NIR leads (23/36); provisional until 'assets fiber profile' is checked.

NIR, diffuse reflection probe · 23/36

- Broad target-signal link; no catalog entry = UNKNOWN, never exclusion. `S01 p4-5` `REF §3,5` `R_Q1_01`
- Preference is soft; a lab interface stays valid for the feasibility experiment. `REF §3` `R_Q3C_05`

Cannot be approved: Reuse of the existing link cannot be approved without the actual cable data; a new-link branch is kept as the alternative. `R_Q3C_08` `S02 p6` `H04`

Recommended methods

#1
23 /36
range [3–32]

NIR · diffuse reflection probe

Confidence: low · ⚠ critical question open

Q1	6 /9	Q1b	4 /9
Q2	5 /9	Q3	8 /9
Q4	1 /3		

#2

22 /36

range [4–32]

MIR · ATR probe

Confidence: low | ⚠ critical question open

Q1	6 /9	Q1b	4 /9
Q2	5 /9	Q3	7 /9
Q4	2 /3		

#3

21 /36

range [3–32]

Raman · Raman probe

Confidence: low | ⚠ critical question open

Q1	4 /9	Q1b	4 /9
Q2	5 /9	Q3	8 /9
Q4	0 /3		

▼ More methods (2)

#4

20 /36

range [3–32]

Fluorescence · fluorimetric probe

Confidence: low | ⚠ critical question open

Q1	4 /9	Q1b	4 /9
Q2	5 /9	Q3	7 /9
Q4	0 /3		

#5
20 /36
range [3–32]

UV-Vis · ATR probe

Confidence: low | ⚠ critical question open

Q1	4 /9	Q1b	4 /9
Q2	5 /9	Q3	7 /9
Q4	0 /3		

Questions for the customer

1. high `ProcessContext · assets · fiber profile`

   Can you send the data sheet of the existing cable (fiber type, length, transmission)?

   ▼ Why

   if fiber_profile=confirmed: leader NIR

   `R_Q3P_03`

2. high `ProcessContext · assets · must reuse`

   Must the existing link/instrument be reused, or is a new link acceptable?

   ▼ Why

   if must_reuse=yes: leader ['NIR|NIR_DIFFUSE_PROBE', 'NIR|NIR_TRANS_PROBE', 'NIR|NIR_DISTANCE']; if must_reuse=no: leader MIR

   `R_Q3C_09`

3. high `Sample · physical form`

   What is the physical form at the measuring point (solution, suspension/emulsion, powder, paste/melt, tissue)?

   ▼ Why

   if physical_form=solution: leader ['NIR|NIR_TRANS_PROBE', 'MIR|MIR_ATR_PROBE', 'NIR|NIR_DIFFUSE_PROBE']; if physical_form=suspension: leader ['NIR|NIR_DIFFUSE_PROBE', 'NIR|NIR_TRANS_PROBE', 'NIR|NIR_DIFFUSE_PROBE']; if physical_form=emulsion: leader ['NIR|NIR_DIFFUSE_PROBE', 'NIR|NIR_TRANS_PROBE', 'NIR|NIR_DIFFUSE_PROBE']

   `R_Q1_M02, R_Q1_N01, R_Q1_N04, R_Q1_R01, R_Q2_02`

To improve accuracy, add the customer's answers to the text above and click Analyze again.

Ruled out (1)

UV_VIS · UV_LENS
This interface collects light emitted by the sample itself; the sample is not a light source. `R_Q2_02`

Decision path · factors of the leading candidate per stage, with source

Q1

MATCH **+2** Broad target-signal link; no catalog entry = UNKNOWN, never exclusion.
`S01 p4-5` `REF §3,5` R_Q1_01

UNKNOWN **0** Separation of the target signal from the interferences of this matrix. MATCH only where an evidence row names the page (selectivity_source); no spectra in the materials = UNKNOWN, never a MISMATCH and never 'no interferences'. `S01 p5,10-11` R_Q1_04

UNKNOWN **0** Generic row: a known matrix background must not cover the target signal. Unknown overlap = UNKNOWN. Replaced by the specific rows R_Q1_R02 (Raman fluorescence) and R_Q1_M03 (MIR water). Soft only; a hard exclusion needs a proven overlap (R_Q1_02). `S01 p5,11` `REF §5` R_Q1_05

UNKNOWN **0** Invalid old calibration does not exclude the physical method; a new calibration plan is mandatory. `REF D01 risks, §5` `S01 p10` R_Q1_06

UNKNOWN **0** Independent validation material must be established, not assumed.
`REF D01/D02 experiment` `S01 p5,10` R_Q1_07

UNKNOWN **0** Particle, density, temperature, texture variation must be in the matrix calibration.
`S01 p10,15,17` `REF D01,D02` R_Q1_N02

Q1b

UNKNOWN **0** LOD/LOQ only from a scoped CapabilityProfile; the materials give none.
`spec 04 §1` `REF §3,5` R_Q1B_02

UNKNOWN **0** Achieved error vs allowed error needs a comparable metric; none in the materials.
`REF D01/D02 experiment` `spec 04` R_Q1B_04

UNKNOWN **0** A laboratory is not a known reference uncertainty. `REF D01,D02` `H01` R_Q1B_05

UNKNOWN **0** End-to-end latency needs a profile; spectral exposure alone is not the latency.
`spec 04 §1` R_Q1B_06

UNKNOWN **0** Both range ends must be inside the calibration/detector scope.
`REF D01,D02` `S01 p16,18` R_Q1B_08

UNKNOWN **0** Qualitative sensitivity rating: context only, never a score. `XLSX sensitivity + notes` R_Q1B_10

Q2

MATCH **+1** Documented Method-interface edge (S01 Table 1). `XLSX` `S01 p4-6` R_Q2_01

UNKNOWN **0** A form missing from the list is UNKNOWN; MISMATCH only for a constructive conflict (solid vs liquid-only cell). `XLSX` `CL10` R_Q2_02

UNKNOWN **0** Excludes only this pair; other interfaces stay. Taking a sample out is R_Q2_10 (one cause, one rule). `S01 p5,10` R_Q2_03

UNKNOWN **0** A surface or local signal is not the bulk composition of a heterogeneous stream without a sampling plan. `S01 p8,13-14,17-18` `REF D08` R_Q2_07

UNKNOWN **0** Particle/bubble/surface variation must be covered by a sampling or averaging plan. Our recommendation to average is not a plan (no MATCH from it). `REF D01,D02` `S01 p15,17` R_Q2_08

Q3

UNKNOWN **0** No numeric limit in the materials. Hot extrusion does not exclude MIR. `S02 p14` `REF D07` R_Q3C_01

UNKNOWN **0** Fouling + contact interface needs a feasible cleaning/maintenance plan; a promise is not a match. `REF D02,D07` R_Q3C_04

MATCH **+2** Preference is soft; a lab interface stays valid for the feasibility experiment. `REF §3` R_Q3C_05

UNKNOWN **0** Do not invent certification. `design comparator` R_Q3C_06

UNKNOWN **0** Long silica cables are documented; no universal allowed length. `S02 p6-10,18` `S01 p15` R_Q3C_07

MATCH **+1** Generic family is a soft concept check; a real cable with proven mismatch is R_Q3P_03. `S02 p6` `REF §5` R_Q3C_08

MATCH **+1** 'There is a cable' does not mean 'must use it'. `REF H01,H03,H04` R_Q3C_09

UNKNOWN **0** Only with the actual cable data; never excludes the whole method. `S02 p6` R_Q3P_03

Risks and assumptions

high The premise that the existing silica cable can be reused cannot be approved: without the actual cable data the reuse remains unproven, and silica links are for NIR/Raman, not MIR ATR, which needs a different fiber type. The existing calibration or link cannot be assumed valid for MIR. What would still meet the goal: place the MIR analyzer at the line and send the result digitally to the other building, or use NIR/Raman over the existing cable once its data are checked.

high The lead is not clearly separated: NIR scores 23/36 vs MIR 22/36, with wide intervals [3, 32] and [4, 32], and Raman 21/36. The confidence is low, and the water range, reference method and fiber profile are unknown, so every conclusion stays conditional.

medium The top-ranked NIR probe is a diffuse reflection probe, but the card lists these mainly for powders; for liquid petroleum a transmission or transflection probe may fit better. Both are scored equally, so the probe choice depends on the physical form at the measuring point.

medium If the water content is in the ppm range, NIR and Raman have limited sensitivity and the water bands may be too small against the petroleum matrix. If it is in the percent range, this risk is smaller.

medium Free or emulsified water produces scattering and drifting baselines. This can disturb the signal, and the probe may not see a representative sample.

medium Petroleum may fluoresce strongly, which could swamp a Raman signal; laser-based Raman in a hazardous area also needs explosion protection. Both are unknown at present.

medium Fouling, wax or deposits on the probe window or ATR crystal, plus unknown temperature and pressure, may exceed the probe or fiber ratings and lower the signal over time.

medium No independent reference method or samples are known, and the knowledge cards are draft and unverified. Calibration transfer between instruments and between batches is untested, especially for NIR.

First feasibility experiment

Find out on real petroleum samples which method (NIR with a transflection or diffuse probe, or MIR with an ATR probe) measures water against your reference method, and whether the existing cable can be reused.

Samples

About 40-60 samples from at least 4 independent batches or runs; the final count is to be agreed with you. — Start from dry or low-water petroleum from the line, spike with known amounts of water over the agreed range, and cover both dissolved and dispersed water if both occur. Vary temperature independently of water content, so that water and temperature do not correlate. Include different product grades or batches, and one set of real line samples with unspiked reference values.

Reference method

Your existing water reference method. If none exists, choose between Karl Fischer titration (coulometric or volumetric) and an agreed lab method, measured on the same samples at the same time.

Methods compared

NIR, MIR, Raman

Interferents

temperature, free or emulsified water (droplets, turbidity), matrix composition between batches, fluorescence of the petroleum, fouling or wax on the probe window

Duration

About 4-6 weeks for the lab study after samples and cable data are available; an in-line campaign at the line would take longer and is planned afterwards.

1. 1. Collect the cable data sheet and measure the existing cable's transmission over the needed length, and check the fluorescence background of a petroleum sample.
2. 2. Agree the water range, accuracy target and reference method with you; define the sample plan.
3. 3. Prepare the spiked and designed samples with reference values; record temperature and spectra with NIR (transflection and diffuse probes) and MIR (ATR probe) on fiber probes; run Raman on a smaller subset if the fluorescence check is acceptable.
4. 4. Build calibrations on at least two calibration batches and test them on independent validation batches (split by batch or run, not by neighbouring spectra of one sample).
5. 5. Compare each method against the reference and against the other methods; also assess the existing cable versus a new link in the same test.
6. 6. Decide go or no-go for a short in-line test at the line, including probe fouling and temperature behaviour.

Success criterion: On independent validation batches, bias against the reference is not significantly different from zero and the prediction error stays within the accuracy agreed with you beforehand; the limit is to be fixed before the study starts.

Parameters and attributes · 38 Facts

Facts

Value · unit · status. Hover a row for the customer's words. Edit → Recalculate.

Fact	Value	Unit	Status
MeasurementTask · measurement goal	quantify		inferred
MeasurementTask · target	water		reported „…“
MeasurementTask · target definition	defined		reported „…“
MeasurementTask · spatial scope	—		unknown
MeasurementTask · sampling plan	—		unknown
MeasurementTask · reference method	—		unknown
MeasurementTask · performance requirements	—		unknown
MeasurementTask · time requirements	—		unknown
MeasurementTask · target range	—		unknown

Fact	Value	Unit	Status
MeasurementTask · business goal	—		unknown
Sample · matrix	petroleum		reported „…“
Sample · phase	liquid		inferred
Sample · physical form	—		unknown
Sample · optical properties · opacity	—		unknown
Sample · optical properties · fluorescence	—		unknown
Sample · heterogeneity	—		unknown
Sample · variability · formulation	—		unknown
Sample · variability · physical	—		unknown
Sample · sample state constraints · contact	—		unknown
Sample · sample state constraints · sampling	—		unknown
Sample · relevant components · fluorophore	—		unknown
Sample · relevant components · absorber	—		unknown
ProcessContext · deployment preference	—		unknown
ProcessContext · temperature range	—		unknown
ProcessContext · temperature c	—	degC	unknown
ProcessContext · operation constraints · fouling	—		unknown
ProcessContext · operation constraints · hazardous	—		unknown
ProcessContext · instrument distance	remote		reported „…“
ProcessContext · assets · fiber	silica_vis_nir		reported „…“

Fact	Value	Unit	Status
ProcessContext · assets · fiber profile	—		unknown
ProcessContext · assets · must reuse	—		unknown
ProcessContext · assets · calibration scope	—		unknown
MeasurementTask · reference method · independent	—		unknown
ProcessContext · assets · method	—		unknown
CustomerRequest · justification	—		unknown
CustomerRequest · proxy allowed	—		unknown
CustomerRequest · asks	approve_concept		reported „…“
CustomerRequest · combination to keep calibration	—		unknown